



ST. XAVIER'S COLLEGE

MAITIGHAR, KATHMANDU, NEPAL

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TITLE - SHEET

Name : Bhavesh Giri

Roll No : 745

Grade : 11

Section : G

Subject : Computer

S.N.	Unit / Chapter	Sub-Topics	☒	Signed	Date	Remarks
1	Computer System	1.1 Components	☐			
		1.2 Bus System	☐			
		1.3 Motherboard	☐			
		1.4 Hardware Interfaces	☐			
2	Number System and Boolean Logic	2.1 Number Conversion	☐			
		2.2 Boolean Logics	☐			
		2.3 Laws of Boolean Algebra	☐			
3	Operating System	3.1 System Configurations	☐			
		3.2 File Explorer Management	☐			
		3.3 Command Line Interface	☐			
4	Application Package	4.1 Word Processing Software	☐			
		4.2 Spreadsheet Software	☐			
		4.3 Spreadsheet Formulas	☐			

		4.4 Presentation Software				
5	Programming Concepts and Logics	5.1 Basic Syntax Framework				
		5.2 C Tokens				
		5.3 Operators & Expressions				
		5.4 Control Structures				
		5.5 Array 1D and 2D				
		5.6 String Handling Functions				
		5.7 Mathematical Problems				
6	Web Technology I	6.1 Document Structure				
		6.2 Core Text elements				
		6.3 Lists & Hyperlinks				
		6.4 Table Structure				
		6.5 Form Elements				
		6.6 Inline & Internal Styling				
		6.7 External Style Sheets				
		6.8 Modern Layout Properties				
7	Multimedia	7.1 Multimedia in Web Page				
8	Information Security and Cyber Law	8.1 Information Security (CIA)				
		8.2 Cryptography				
		8.3 Security Measures				
		8.4 Cyber Crimes				
		8.5 Cyber Laws				

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LAB/WRITTEN ASSIGNMENT NUMBER: 4.2

“Spreadsheet Software”

Submitted By	Submitted To	Signature
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Part A: Theory

Answer the following questions

- 1) Define a cell in Microsoft Excel.

A cell in Microsoft Excel is the basic unit of the spreadsheet, which is defined as the intersection of a row and a column.

- 2) Draw and label the blueprint (interface) of Microsoft Excel, including the following components:

- o Title Bar
- o Quick Access Toolbar
- o Ribbon
- o Name Box
- o Formula Bar
- o Column Headings
- o Row Headings
- o Active Cell
- o Worksheet Area
- o Sheet Tabs
- o Status Bar
- o Zoom Slider

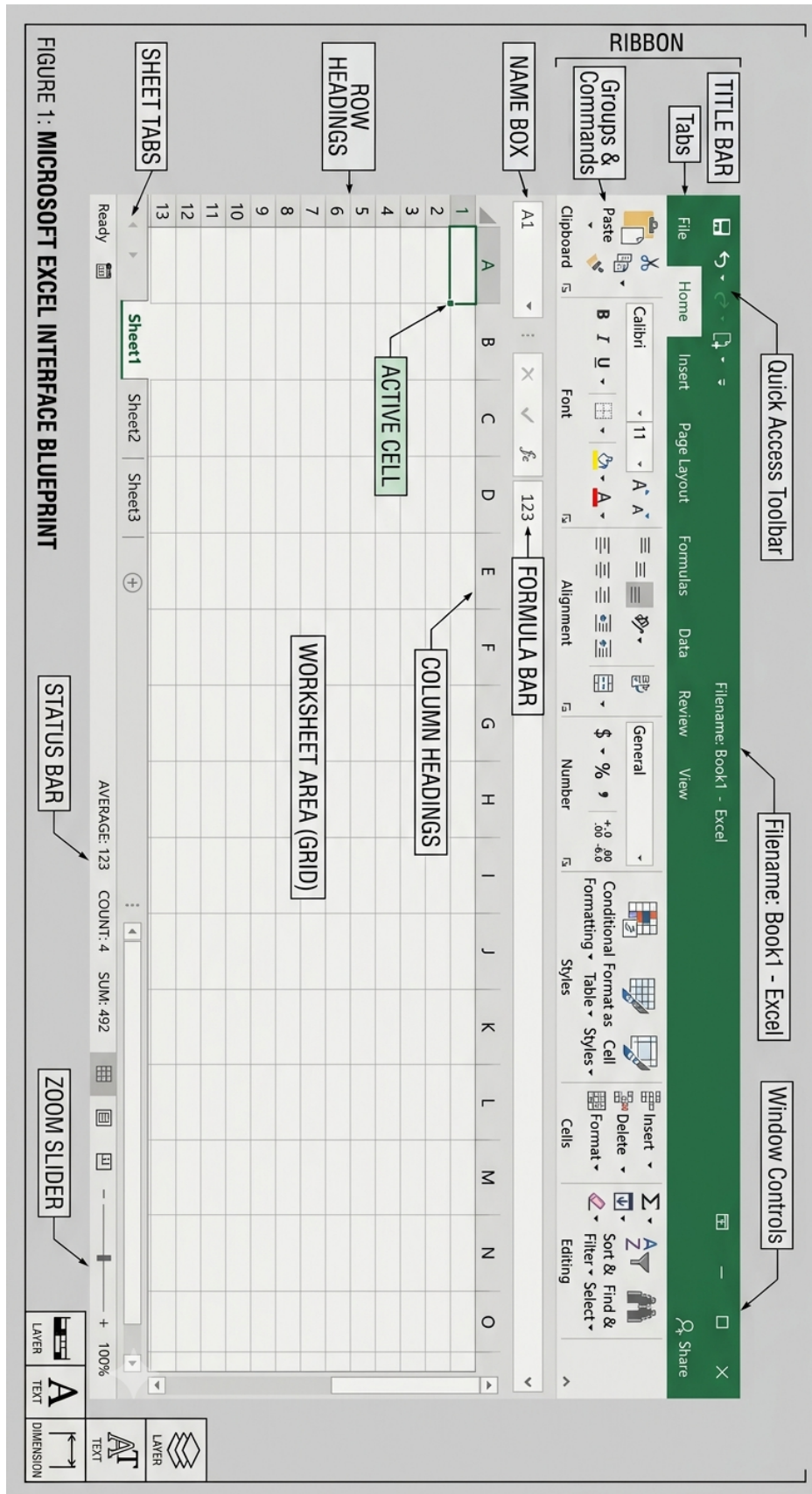


FIGURE 1: MICROSOFT EXCEL INTERFACE BLUEPRINT

Figure 4.1:Structural blueprint of Microsoft Excel Interface

3) Define Relative Cell Reference with one example.

A relative cell reference is a cell address that automatically adjusts when a formula is copied or moved to a different location, based on the relative position of the cells.

For example,

In cell C1, you enter the formula $C1=A1+B1$. If you copy this formula to cell C2, it automatically adjusts to $C2=A2+B2$.

4) Define Mixed Cell Reference with one example.

A mixed cell reference in Excel is a formula reference that combines absolute and relative components where either the column or the row is fixed while the other adjusts when the formula is copied.

For example,

$\$A1$ is a mixed reference where the column (A) is absolute and remains constant, while the row (1) is relative and changes as the formula is dragged down or across cells.

5) Define Absolute Cell Reference with one example.

An absolute cell reference is a cell address that remains fixed and does not change when a formula is copied or dragged to other cells.

For example,

It is identified by the dollar sign (\$) preceding both the column letter and row number, $\$A\1

6) Define VLOOKUP.

VLOOKUP (Vertical Lookup) is an Excel function that searches for a specific value in the first column of a table or range and returns a corresponding value from the same row in a different column.

7) Write the syntax of VLOOKUP and explain each argument with an example.

The syntax for the VLOOKUP function in Excel is:

$=VLOOKUP(\text{lookup_value}, \text{table_array}, \text{col_index_num}, [\text{range_lookup}])$

lookup_value: The value you want to search for, which must be in the first column of your specified table range.

Example: If you want to find the price of a product with ID 101, your lookup_value is 101.

table_array: The range of cells containing your data. The first column of this range must contain the **lookup_value**.

Example: If your data spans columns A to C from rows 2 to 10, use A2:C10.

col_index_num: The column number (starting from 1 for the left-most column of the **table_array**) from which to return the result.

Example: If you want the value from the third column of your table array, you enter 3.

[range_lookup]: Optional. Specifies whether to find an exact match or an approximate match. Use FALSE (or 0) for an exact match. Use TRUE (or 1) for an approximate match (requires the first column to be sorted).

Example: =VLOOKUP(101, A2:C10, 3, FALSE) looks for ID 101 and returns the exact value from the 3rd column.

Part B: Practical

Sheet 1: Student Mark sheet Calculation

Office Update To keep up-to-date with security updates, fixes, and improvements, choose Check for Updates.

Roll no	Name	Physics Th	Physics Pr	Chemistry Th	Chemistry Pr	Computer Th	Computer Pr	Nepali Th	Nepali Pr	English Th	English Pr	Math Th	Math Pr	Total	Remark	Percentage	Grade	Rank
1	Bhavesh Giri	74	24	74	24	49	49	74	24	74	24	74	24	588	PASS	98.8	A+	1
2	Bhawana Basnet	40	16	40	16	30	30	40	16	40	16	40	16	340	PASS	56.666667	C+	6
3	Rajesh Hamal	60	20	70	20	48	48	70	24	65	22	66	20	533	PASS	88.833333	A	3
4	Shah Rukh Khan	65	20	74	22	12	12	16	45	45	70	24	54	471	FAIL	NG	NG	NG
5	Lana Del Rey	65	22	70	20	16	8	16	26	50	22	51	22	388	FAIL	NG	NG	NG
6	Michael Jackson	64	25	75	16	28	25	33	22	56	21	65	21	451	PASS	75.166667	B+	5
7	Nachal KC	60	23	66	15	30	26	22	22	23	20	52	16	375	FAIL	NG	NG	NG
8	Justin Kar	62	20	65	6	33	22	45	23	54	25	65	10	430	FAIL	NG	NG	NG
9	Shweta Adhikari	72	24	72	16	40	40	70	24	66	23	57	24	528	PASS	88	A	4
10	Aradhya Tamrakar	71	22	71	17	41	39	69	21	71	22	74	22	543	PASS	90.5	A+	2

BHAVESH GIRI | Roll No.: 745

Figure 4.2: Screenshots of Sheet 1: Student Mark sheet Calculation

Excel formula

1) Total

Roll no	Name	Physics Th	Physics Pr	Chemistry Th	Chemistry Pr	Computer Th	Computer Pr	Nepali Th	Nepali Pr	English Th	English Pr	Math Th	Math Pr	Total	Remark	Percentage	Grade	Rank
1																		

2) Percentage

Roll no	Name	Physics Th	Physics Pr	Chemistry Th	Chemistry Pr	Computer Th	Computer Pr	Nepali Th	Nepali Pr	English Th	English Pr	Math Th	Math Pr	Total	Remark	Percentage	Grade	Rank
1																		

3) Grade

Roll no	Name	Physics Th	Physics Pr	Chemistry Th	Chemistry Pr	Computer Th	Computer Pr	Nepali Th	Nepali Pr	English Th	English Pr	Math Th	Math Pr	Total	Remark	Percentage	Grade	Rank
1																		

4) Remark

Roll no	Name	Physics Th	Physics Pr	Chemistry Th	Chemistry Pr	Computer Th	Computer Pr	Nepali Th	Nepali Pr	English Th	English Pr	Math Th	Math Pr	Total	Remark	Percentage	Grade	Rank
1																		

5) Rank

Roll no	Name	Physics Th	Physics Pr	Chemistry Th	Chemistry Pr	Computer Th	Computer Pr	Nepali Th	Nepali Pr	English Th	English Pr	Math Th	Math Pr	Total	Remark	Percentage	Grade	Rank
1																		

Sheet 2: VLOOKUP-Based Student Result

ST. XAVIER'S COLLEGE
Maitighar, Kathmandu
Dedicated to Excellence, Leadership, Service

ANNUAL REPORT CARD

Roll No.: 5
Student Name: Lara Dal Ray
Section: G

S.N.	Subject	Total Theory Marks	Total Practical Marks	Marks Obtained in Theory	Marks Obtained in Practical
1	English	75	25	74	24
2	Nepali	75	25	74	24
3	Computer	50	50	49	49
4	Physics	75	25	74	24
5	Chemistry	75	25	74	24
6	Math	75	25	74	24
Total		600	200	588	200
Remarks		PASS			
Percentage		98			
Grade		A+			
Rank		1			

Exam Coordinator: Mr. Lal Mani Joshi
Principal: Fr. Dr. Augustine Thomas, S.J.
Vice-Principal: Mr. Gehendra Bahadur Chand

Figure 4.3: Screenshots of Sheet 2: VLOOKUP-Based Student Result

ST. XAVIER'S COLLEGE
Maitighar, Kathmandu
Dedicated to Excellence, Leadership, Service

ANNUAL REPORT CARD

Roll No.: 1
Student Name: Bhavesh Giri
Section: G

S.N.	Subject	Total Theory Marks	Total Practical Marks	Marks Obtained in Theory	Marks Obtained in Practical
1	English	75	25	74	24
2	Nepali	75	25	74	24
3	Computer	50	50	49	49
4	Physics	75	25	74	24
5	Chemistry	75	25	74	24
6	Math	75	25	74	24
Total		600	200	588	200
Remarks		PASS			
Percentage		98			
Grade		A+			
Rank		1			

Exam Coordinator: Mr. Lal Mani Joshi
Principal: Fr. Dr. Augustine Thomas, S.J.
Vice-Principal: Mr. Gehendra Bahadur Chand

Figure 4.4: Marksheet Roll no. 1

ST.XAVIER'S COLLEGE				
Maitighar, Kathmandu				
Dedicated to Excellence, Leadership, Service				
		ANNUAL REPORT CARD		
Roll No.: 2 Student Name: Bhawana Basnet Section: G				
S.N.	Subject	Total Theory Marks	Total Practical Marks	Marks Obtained in Theory
1	English	75	25	40
2	Nepali	75	25	40
3	Computer	50	50	30
4	Physics	75	25	40
5	Chemistry	75	25	40
6	Math	75	25	40
Total		600		340
		Remarks		PASS
		Percentage		56.66666667
		Grade		C+
		Rank		6
Exam Coordinator Mr. Lal Mani Joshi		Principle Fr. Dr. Augustine Thomas, S.J.		Vice-Principle Mr. Gehendra Bahadur Chand

Figure 4.5: Marksheet Roll no. 2

ST.XAVIER'S COLLEGE				
Maitighar, Kathmandu				
Dedicated to Excellence, Leadership, Service				
		ANNUAL REPORT CARD		
Roll No.: 3 Student Name: Rajesh Hamal Section: G				
S.N.	Subject	Total Theory Marks	Total Practical Marks	Marks Obtained in Theory
1	English	75	25	65
2	Nepali	75	25	70
3	Computer	50	50	48
4	Physics	75	25	60
5	Chemistry	75	25	70
6	Math	75	25	66
Total		600		533
		Remarks		PASS
		Percentage		88.83333333
		Grade		A
		Rank		3
Exam Coordinator Mr. Lal Mani Joshi		Principle Fr. Dr. Augustine Thomas, S.J.		Vice-Principle Mr. Gehendra Bahadur Chand

Figure 4.6: Marksheet Roll no. 3

ST.XAVIER'S COLLEGE				
Maitighar, Kathmandu				
Dedicated to Excellence, Leadership, Service				
		ANNUAL REPORT CARD		
Roll No.: 4 Student Name: Shah Rukh Khan Section: G				
S.N.	Subject	Total Theory Marks	Total Practical Marks	Marks Obtained in Theory / Marks Obtained in Practical
1	English	75	25	70 / 24
2	Nepali	75	25	45 / 45
3	Computer	50	50	12 / 16
4	Physics	75	25	65 / 20
5	Chemistry	75	25	74 / 22
6	Math	75	25	54 / 24
Total		600		471
		Remarks		FAIL
		Percentage		NG
		Grade		NG
		Rank		NG
Exam Coordinator Mr. Lal Mani Joshi		Principle Fr. Dr. Augustine Thomas, S.J.		Vice-Principle Mr. Gehendra Bahadur Chand

Figure 4.7: Marksheet Roll no. 4

ST.XAVIER'S COLLEGE				
Maitighar, Kathmandu				
Dedicated to Excellence, Leadership, Service				
		ANNUAL REPORT CARD		
Roll No.: 5 Student Name: Lana Del Rey Section: G				
S.N.	Subject	Total Theory Marks	Total Practical Marks	Marks Obtained in Theory / Marks Obtained in Practical
1	English	75	25	50 / 22
2	Nepali	75	25	16 / 26
3	Computer	50	50	16 / 8
4	Physics	75	25	65 / 22
5	Chemistry	75	25	70 / 20
6	Math	75	25	51 / 22
Total		600		388
		Remarks		FAIL
		Percentage		NG
		Grade		NG
		Rank		NG
Exam Coordinator Mr. Lal Mani Joshi		Principle Fr. Dr. Augustine Thomas, S.J.		Vice-Principle Mr. Gehendra Bahadur Chand

Figure 4.8: Marksheet Roll no. 5

Final Sections

Lab activity reflections

During this lab session I learned how Excel handles cell references in formulas. I practiced cell references, which change automatically when I copy a formula to a new cell and absolute cell references, which stay fixed using the dollar sign (\$) so the formula always points to the same cell. This helped me understand why some formulas gave results when copied without locking the reference.

I also learned the VLOOKUP function. I used it to search for a value in one column and return a matching value from another column in the row. At first, I struggled with the syntax. Got #N/A errors when my lookup value did not match exactly. After checking the table array and setting the range lookup to FALSE for a match my formula worked correctly.

The useful part of this lab was preparing a mark sheet. I entered student names and marks for subjects then used formulas to calculate totals, percentages and grades. I used VLOOKUP to pull student details from a data sheet into the mark sheet, which saved time and reduced manual entry errors.

This lab activity improved my confidence, in using Excel formulas for tasks. I now understand how relative and absolute references work with VLOOKUP to build an accurate and efficient mark sheet.

Conclusion

This lab activity strengthened my practical skills in Microsoft Excel, particularly in using cell references and lookup functions to solve real world data problems. I understood the difference between relative and absolute references and applied them correctly in formulas. I also learned to use VLOOKUP for accurate data retrieval across sheets, which reduced manual errors and saved time. By preparing a complete mark sheet with calculated totals, percentages, and grades, I gained hands-on experience in organizing and processing data efficiently. This assignment improved my confidence in using Excel as a tool for academic and professional record management.